

Volatility Is Hard to Beat: A Survivorship-Free Benchmark of Downside-Risk Metrics and the Limits of LLM Metric Discovery

Working paper — draft, revision 2 (2026-07-13). Target: ACM ICAIF (short paper). All results reproduce from `src/` and `outputs/`; every number traces to `context/findings.md`. Published-metric implementations are daily-data adaptations, not exact replications, and several published metrics are **repurposed** here as drawdown forecasters (see §3, §6). References verified against publisher records.

Abstract

Machine-driven quantitative research — LLM agents that write and test their own signals — makes it trivially cheap to search a vast space of risk metrics. We show, on a downside-risk case study, that this cheap search is a data-snooping engine that a locked holdout is **required** to expose, and that the metrics it (and the published literature) produce do not reliably beat plain volatility. Concretely: (i) an autonomous LLM autoresearch loop and a 20,000-config directed search inflate **validation** skill monotonically with search intensity while the **locked-test** score plateaus at the volatility baseline; validation-selected metrics beat volatility out-of-sample only ~43–54% of the time (a coin flip). (ii) Benchmarking ten metrics (eight published, two classical) across **five test beds** spanning calm and crash regimes in **crypto (survivorship-free, dead coins retained) and equity**, with **bootstrap confidence intervals** on every comparison: the only statistically significant wins over volatility are two volatility-**variants** (downside deviation, VaR) on a single crypto-bull bed, and they **reverse** on equity; the genuinely different, return-**pricing** metrics (semibetas, downside beta, tail dependence) — repurposed here as drawdown forecasters, a use their authors did not claim — are significantly **worse** than volatility. (iii) The one robust positive: volatility is a strong survivorship-free **blow-up** predictor (ROC-AUC up to 0.95; 2.5× top-decile lift), which a gradient-boosting model fails to improve on. The takeaway is a methodological one for AI-in-finance: automated metric discovery demands anti-snooping guardrails — a locked, ideally survivorship-free, holdout is not optional.

1. Introduction

LLM agents are now wired into quantitative-research loops that propose, code, and score candidate signals autonomously. This collapses the cost of specification search to near zero — and thereby makes the classical data-snooping problem (White, 2000; Harvey, Liu & Zhu, 2016) acute in a new way: an agent can try thousands of metric variants unsupervised. We ask what discipline is required for such loops to produce trustworthy signals, using downside-risk forecasting as a concrete, well-populated case study.

The case study is motivated by auditing Viole & Nawrocki (2016): their partial-moment "predictive" metric proved to have an inert autocorrelation term and a survivorship-inflated edge. That raised the fair question — on a level field (survivorship-free data, a locked holdout, multiple regimes, significance tests) — does **any** downside-risk metric, published or machine-discovered, beat plain volatility at forecasting tail risk?

Contributions. (1) Evidence that autonomous metric discovery inflates validation but not a locked-test score (search-induced overfitting), and that this is exposed only by a locked, survivorship-free holdout — a cautionary result for LLM-driven quant research, with a released guarded harness. (2) A significance-tested, multi-regime, survivorship-free benchmark of ten downside-risk metrics showing none consistently beats volatility. (3) A useful positive: volatility as a survivorship-free blow-up predictor. We deliberately report a **benchmark and a cautionary tale**, not a new metric — our attempts to build one, including the discovery loop, failed to beat volatility once the guardrails were in place.

2. Test beds and protocol

Five test beds, calm and crash, two asset classes. Crypto (survivorship-free): a CoinMarketCap daily panel (2013–2018) that retains crashed/delisted coins, so a point-in-time top-100-by-market-cap universe includes assets that later crater; we use three regimes — **2016 (calm), 2017 (bull), 2018 (crash)**. **Equity:** S&P 500 constituents (yfinance), two regimes — **1994–1999 (calm bull, n=66**

months) and **2005-2009 (GFC, n=54)**. The equity beds are survivorship-**biased** by construction (free equity retains almost no true deaths — itself a finding); they test only cross-asset generality of the ranking, **not** the survivorship claim, which rests on the crypto beds.

Skill measure and significance. From a 180-day trailing window we forecast 90-day forward maximum drawdown (delisting scored -100%); skill is the per-formation-date cross-sectional Spearman correlation, averaged over the bed. Crucially, we place a **paired block-bootstrap 95% CI on the difference (metric – volatility)** for every comparison; a metric "beats volatility" only if that CI excludes zero. (This corrects a bare point-estimate ranking.)

Machine discovery (§4.2) uses a strict train/validation/**locked-test** split: the search optimizes validation; the test is scored once, after the metric is frozen.

3. Metrics

We group ten metrics by what they were **designed** to do, because most were not built to forecast drawdown and are **repurposed** here: - **Volatility family (risk magnitude):** volatility; downside deviation; 5% VaR; 5% ES (Atilgan et al., 2020). These are near-transforms of trailing dispersion. - **Return-pricing / tail-comovement measures (repurposed):** Viole-Nawrocki UPM/LPM ratio; Bollerslev-Patton-Quaedvlieg negative semibeta; Ang-Chen-Xing / Levi-Welch downside beta; Chabi-Yo et al. lower-tail dependence; Farago-Tédongap volatility-downside. These were built to price the cross-section of **returns** or premia, not to rank an asset's forward drawdown.

We drop Kelly-Jiang tail risk from the cross-sectional table: it is an **aggregate** time-series estimator (pooled Hill exponent predicting the market premium), not a per-asset measure, and does not have a faithful per-asset analogue. All implementations are daily adaptations (an appendix documents each estimator and its fidelity trade-off); results for the pricing measures should be read as "does this construct, operationalized comparably, forecast drawdown," not as a verdict on the originals' pricing claims.

4. Results

4.1 Automated discovery inflates validation, not the locked test (lead result)

We built a Karpathy-style autoresearch loop: an LLM agent freely rewrites a single `metric.py`; a frozen scorer evaluates it on validation; improvements are kept; the discovery is scored once on the locked test. A 20,000-config directed search makes the mechanism unambiguous (Table 1): the best **validation** score rises monotonically with search intensity while the **locked-test** score plateaus at the volatility baseline.

Search intensity N	Best validation	Its locked test
1	0.27	0.31
100	0.35	0.37
20,000	0.356	0.373 (vol baseline 0.376)

Table 1. Best-of-N. Validation inflates with search; locked test does not. Validation-selected metrics beat volatility on the locked test only **~43-54%** of the time — a coin flip (42.7% in the 20k directed sweep; 53.6% in an independent multi-run study of shorter searches). An open-ended 50-experiment LLM loop reached a locked-test 0.42 (a real but small gain that did not survive as a **significant** improvement and was not reproduced by an independent greedy search), reinforcing that searching harder buys nothing reliable out of sample.

Novelty (honest). The **statistical** phenomenon is classical data-snooping (White, 2000). Our contribution is that it now occurs via an **autonomous agent** proposing thousands of code-level metrics unsupervised, collapsing search cost to near zero — so the locked-holdout guardrail becomes non-optional. Notably, a **non-LLM** random-metric sweep reproduces the same plateau, which is the point: the danger is the **cheap search**, not the language model per se, and the remedy is the holdout.

4.2 No downside-risk metric consistently beats volatility (significance-tested)

Table 2 reports out-of-sample drawdown-forecast rank correlations across the five test beds, with significance vs volatility from paired block-bootstrap CIs.

Metric	crypto-16 calm	crypto-17 bull	crypto-18 crash	equity 94-99	equity 05-09
volatility (baseline)	0.623	0.355	0.376	0.606	0.564
downside deviation	0.615	0.379*	0.410	0.582 †	0.535 †
VaR5	0.611 †	0.400*	0.362	0.580 †	0.540 †
ES5 (Atilgan)	0.623	0.351	0.327 †	0.581 †	0.522 †
semibeta (BPQ)	0.400 †	0.257 †	0.309 †	0.487 †	0.462 †
downside beta (ACX/LW)	0.014 †	0.093 †	0.151 †	0.242 †	0.344 †
lower-tail dep. (Chabi-Yo)	−0.376 †	−0.154 †	−0.285 †	−0.065 †	−0.046 †
vol-downside (Farago-Téd.)	−0.055 †	−0.105 †	−0.050 †	−0.243 †	−0.321 †
UPM/LPM (Viole-Nawrocki)	−0.549 †	−0.276 †	−0.346 †	−0.426 †	−0.424 †

Table 2. OOS drawdown-forecast Spearman. * = significantly beats volatility (95% block-bootstrap CI on the per-date difference excludes 0); † = significantly worse; unmarked = not distinguishable from volatility. Volatility's own skill is highest in the calm crypto regime (0.623).

The **only** significant wins over volatility, on any bed, are two volatility-variants (downside deviation, VaR5) on the single crypto-bull bed — and they **reverse on equity**, where volatility significantly beats them. Every genuinely different (pricing) metric is significantly **worse** than volatility, as expected when a return-pricing measure is repurposed to forecast drawdown. There is thus **no metric that consistently or generalizably beats volatility** across regimes and asset classes. Note this is a narrower and more defensible claim than "nothing beats volatility": small, regime-specific edges exist for volatility's own variants but do not survive a change of asset class.

4.3 The useful positive: volatility predicts blow-ups

Forecasting **which** assets crater (forward return $\leq -80\%$) has real signal. On the locked crypto test, plain volatility attains ROC-AUC 0.68 (0.95 at a -90% threshold) and a $2.5\times$ top-decile lift; a gradient-boosting model **overfits and loses to volatility** (AUC 0.59). This is the paper's most decision-relevant result — though it, too, is a ranking measure, not a cost-aware P&L (see §6).

4.4 Robustness (ablations)

The pattern is robust to forecast horizon (30, 90 days) and to the blow-up threshold ($-50/-70/-90\%$: volatility is the best blow-up predictor at every threshold). A 180-day horizon is infeasible on this data (forward windows exceed the 2018 panel end).

5. Discussion

Two disciplines the source literature usually omits — a survivorship-free universe and a locked holdout with significance tests — flip the picture from "we found a better metric" to "nothing reliably beats volatility, and automated search only manufactures validation mirages." The autoresearch result is the transferable lesson for AI-in-finance: as LLM agents make signal search nearly free and unsupervised, a locked, ideally survivorship-free holdout is the minimum guardrail before trusting a discovered metric. As an integrity note, an earlier version of this project advanced a "survivorship masks downside value" thesis that a hardened significance test refuted; we retracted it, and this paper reports only the claims that survived stress-testing.

6. Limitations

The skill measure is a rank correlation, **not** a cost-aware / tradable P&L; a decile-sorted panel exists in the artifact but a turnover/cost-adjusted result is future work, and conclusions are about forecast ordering, not net value. Published pricing metrics are **repurposed** as drawdown forecasters and implemented as daily adaptations (e.g., lower-tail dependence as a quintile co-exceedance) — their negative results are about this repurposing, not the originals' pricing claims. The equity beds are survivorship-biased and cannot speak to the survivorship thesis (which rests on crypto). Five test beds are not all markets. These bound generality, but the direction — volatility is hard to beat, and search doesn't help — is consistent across every bed. Finally, we evaluate raw cross-sectional drawdown-forecast rank correlation; an earlier exploration found ES beats volatility on a partial-correlation (incremental-beyond-vol) criterion, but that stricter/different task is superseded here by the head-to-head ranking, on which ES is not distinguishable from (or worse than) volatility.

7. Conclusion

On a survivorship-free, multi-regime, significance-tested benchmark, no downside-risk metric — published or machine-discovered — consistently beats plain volatility at forecasting tail risk; the only significant wins are small, regime-specific volatility variants that reverse across asset classes. Volatility is a strong survivorship-free blow-up predictor. And automated, LLM-driven metric discovery inflates validation while a locked holdout stays flat — a cautionary result for the practice of AI-in-finance.

Appendix — reproducibility

src/autoresearch/: frozen prepare_data.py, agent-edited metric.py, frozen run_metric.py/evaluate.py, locked final_eval.py, search_loop.py/ multi_run_study.py/sweep.py (discovery), run_blowup_predictor.py, run_multi_testbed_v2.py (significance + calm regimes), run_ablations.py, analysis.ipynb. Data: CoinMarketCap crypto dump; yfinance S&P 500 + ^GSPC; fja05680/sp500 point-in-time membership. Results in outputs/; decision trail in context/.

References

Ang, Chen & Xing (2006, RFS); Ang & Chua (1979); Atilgan, Bali, Demirtas & Gunaydin (2020, JFE); Bawa (1975); Bollerslev, Patton & Quaedvlieg (2022, JFE); Brown, Goetzmann, Ibbotson & Ross (1992); Chabi-Yo, Ruenzi & Weigert (2018, JFQA); Farago & Tédongap (2018, JFE); Fishburn (1977); Harvey, Liu & Zhu (2016, RFS); Kelly & Jiang (2014, RFS); Levi & Welch (2020, RFS); Novy-Marx & Velikov (2016, RFS); Sortino & van der Meer (1991); Viole & Nawrocki (2016, JMF); Welch & Goyal (2008, RFS); White (2000, Econometrica).